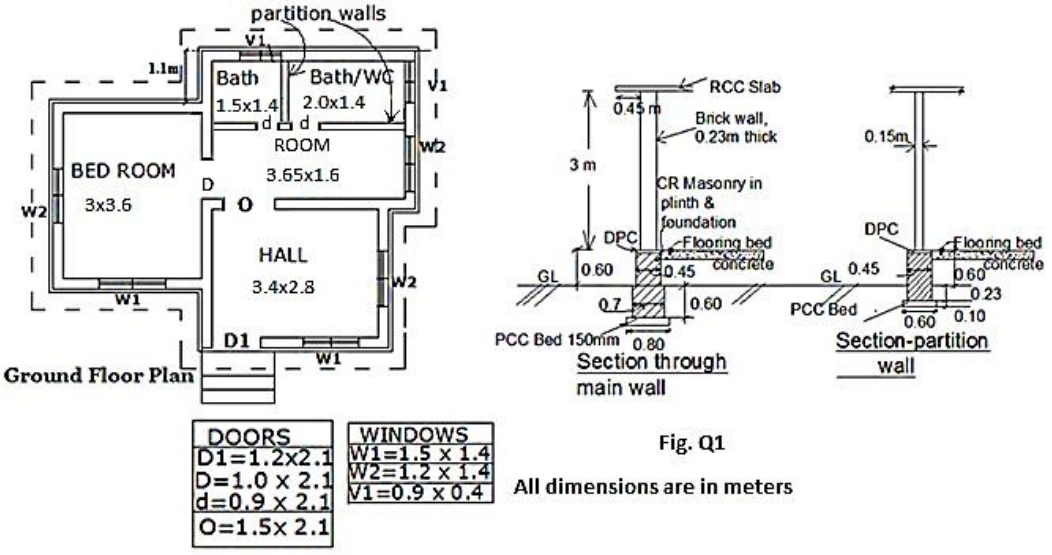
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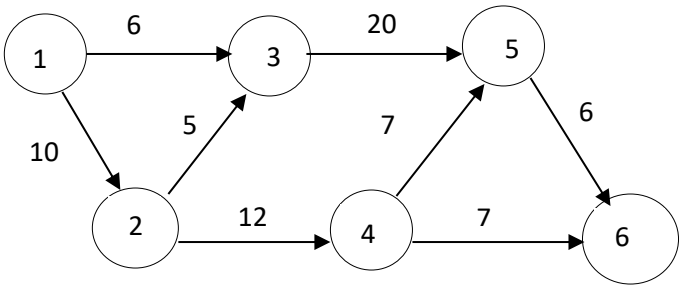
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**C**
Course Name	**ESTIMATION COSTING AND PROJECT MANAGEMENT**	**Course Code**	**CIE-3123**
Semester	**V**	**Max Marks**	**50**
Date	**26-11-2024**	**Duration**	**3 Hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Mark	NBA		BL
			CO	PO	
1A	The accompanying sketch (Fig. Q1) shows the plan of a residential building and a section through the walls. Estimate the quantities of following items of work. (i) Stone masonry in foundation using centerline method (ii) Internal plastering quantity as per IS 1200  **Fig. Q1** **All dimensions are in meters**	**6**	**2**	1,2, 3,6, 11	**3**
1B	Prepare a comprehensive specification for a first class brickwork item of work, incorporating considerations such as material choice, application methods, quality control measures and protection of work.	**4**	1	1, 11, 12	**3**

<table>
<tr><td>2A</td><td>Analyse the unit rate for 12mm thick plastering in cement mortar (1:6) on new brick wall.</td><td>5</td><td>2</td><td>1,2
3,6</td><td>4</td></tr>
<tr><td>2B</td><td>Prepare a Gantt chart for the project with activities, durations, and dependencies as shown in the table below. Additionally calculate and display the overall project progress by the end of the 14th week.
<table>
<tr><th>ID</th><th>Activity</th><th>Duration (Weeks)</th><th>Interdependencies</th></tr>
<tr><td>A</td><td>Site Preparation</td><td>2</td><td>-</td></tr>
<tr><td>B</td><td>Foundation Work</td><td>3</td><td>A</td></tr>
<tr><td>C</td><td>Structural Framework</td><td>4</td><td>B</td></tr>
<tr><td>D</td><td>Electrical and Plumbing</td><td>5</td><td>C</td></tr>
<tr><td>E</td><td>Roofing</td><td>4</td><td>C</td></tr>
<tr><td>F</td><td>Installing wall panels</td><td>3</td><td>C, D</td></tr>
<tr><td>G</td><td>Interior Finishing</td><td>6</td><td>D,E,F</td></tr>
<tr><td>H</td><td>Final Inspection</td><td>2</td><td>G</td></tr>
</table></td><td>5</td><td>3</td><td>1,2
,3,
6,
11</td><td>3</td></tr>
<tr><td>3A</td><td>Following are the relationships between different activities making up a project. Construct a neat AOA and AON network for the interrelationships mentioned below. Use Fulkerson’s rule for numbering AOA network.
<table>
<tr><th>Activity</th><th>Immediate Predecessor(s)</th><th>Duration</th></tr>
<tr><td>A</td><td>-</td><td>3</td></tr>
<tr><td>B</td><td>-</td><td>3</td></tr>
<tr><td>C</td><td>A</td><td>4</td></tr>
<tr><td>D</td><td>B</td><td>5</td></tr>
<tr><td>E</td><td>A, D</td><td>6</td></tr>
<tr><td>F</td><td>B</td><td>5</td></tr>
<tr><td>G</td><td>C, E</td><td>8</td></tr>
<tr><td>H</td><td>F, G</td><td>9</td></tr>
</table></td><td>6</td><td>4</td><td>1,2
,3,
6,
11</td><td>3</td></tr>
<tr><td>3B</td><td>With a graphical representation, explain the significance of direct, indirect and total cost curves associated with a project.</td><td>4</td><td>4</td><td>1,2
,3,
5,9</td><td>2</td></tr>
<tr><td>4A</td><td>Estimate the total float using activity time estimates by critical path method (table approach) for the activity relationship shown in the table below. Also show the critical path on the network identified through float calculations.
<table>
<tr><td>Activity</td><td>1-2</td><td>1-3</td><td>2-4</td><td>3-4</td><td>3-5</td><td>4-5</td><td>4-6</td><td>5-6</td></tr>
<tr><td>Duration</td><td>4</td><td>6</td><td>3</td><td>0</td><td>7</td><td>5</td><td>4</td><td>5</td></tr>
</table></td><td>5</td><td>4</td><td>1,2
,3,
6,
11</td><td>3</td></tr>
</table>

<table>
<tr>
<td>4B</td>
<td>Determine the minimum cost and optimum duration for the activity relationship shown in the table below. Take indirect cost of Rs. 1000 per week.
<table>
<tr><th>Activity</th><th>Normal duration (week)</th><th>Crash duration (week)</th><th>Normal cost (Rs.)</th><th>Crash cost (Rs.)</th></tr>
<tr><td>1-2</td><td>5</td><td>4</td><td>2000</td><td>3000</td></tr>
<tr><td>1-4</td><td>7</td><td>5</td><td>4000</td><td>5700</td></tr>
<tr><td>2-3</td><td>5</td><td>3</td><td>5000</td><td>6500</td></tr>
<tr><td>2-5</td><td>6</td><td>4</td><td>3000</td><td>6000</td></tr>
<tr><td>3-5</td><td>6</td><td>4</td><td>6000</td><td>7000</td></tr>
<tr><td>4-5</td><td>4</td><td>2</td><td>4000</td><td>6000</td></tr>
</table>
</td>
<td>5</td><td>4</td><td>1,2,3,6,11</td><td>4</td>
</tr>
<tr>
<td>5A</td>
<td>Explain the significance of Earned value management (EVM) with regard to various parameters used to analyse cost performance and schedule performance using a typical cumulative cost curve (S-curve).</td>
<td>4</td><td>5</td><td>1,2,3,5,9</td><td>2</td>
</tr>
<tr>
<td>5B</td>
<td>The network of a project is shown in Fig. 5B. The network has been updated after 14 days of its commencement. At the time of updating following conditions were observed.
1. Activity 1-2 and 1-3 have been completed as originally planned
2. Activity 2-4 is in progress and will need 16 days for its completion
3. Activity 4-5 has is in progress and is expected to be completed in 10 days
4. Activity 4-6 can be completed in 5 days instead of 7 days.
5. All the remaining activities are expected to complete as per the original duration.
Update the network using tabular form and compare the critical path of the old and updated network and also compute the delay in the project after updating.
 

</td>
<td>6</td><td>5</td><td>1,2,3,6,11</td><td>3</td>
</tr>
</table>

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO	Total Marks
1	***4***
2	***11***
3	***5***
4	***20***
5	***10***

BL	Total Marks
2	**8**
3	**32**
4	**10**
5	**-**